

Medium Access Control

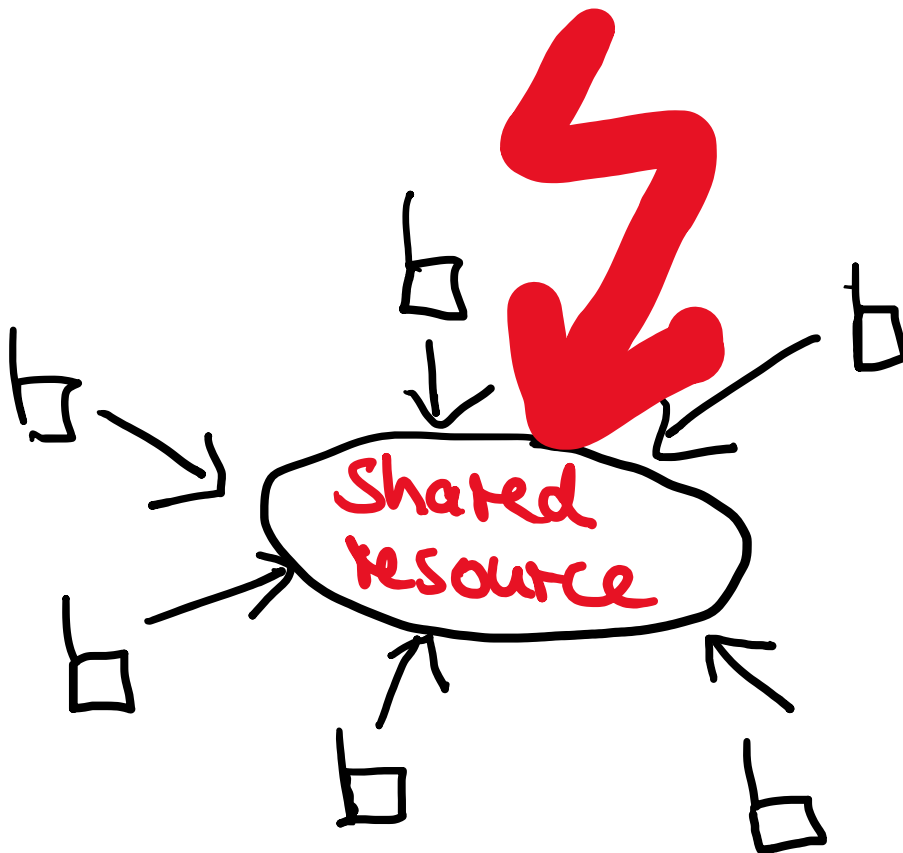
Advanced Mobile Communication Networks



The Medium Access Problem

The problem:

Multiple senders compete for a common, shared resource, i.e. want to use the same frequency, at same time, in the same space



The question:

How can we resolve this conflict?

Basics of Wireless Access Control

Questions to be answered:

- What is the responsibility of the Media Access Control sublayer in communication systems?
- What are the new problems wrt MAC in wireless systems?
- What are the causes of the problems in wireless systems?
- Why does CSMA/CD known from Ethernet fail?
- What is the hidden terminal problem? When does it happen? What happens?
- What is the exposed terminal problem? When does it happen? What happens?
- What is the near far problem? What happens there?
- What are the dimensions in which we can separate the access?

Methods for multiple access

FDMA (Frequency Division Multiple Access)

- assign a certain frequency to a transmission between a sender and a receiver

TDMA (Time Division Multiple Access)

- assign the fixed sending frequency to a transmission between a sender and a receiver for a limited amount of time, e.g. a single MAC frame or an even smaller time slot

CDMA (Code Division Multiple Access)

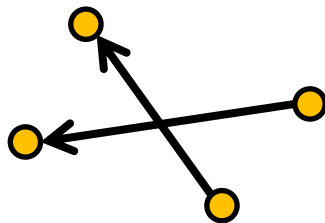
- assign a code to a transmission between a sender and a receiver

SDMA (Space Division Multiple Access)

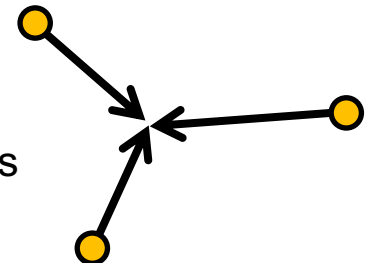
- assign a transmission between a sender and receiver to a certain segment of a cell or a beam

Multiplexing vs. Medium Access: The multiplexing schemes presented previously for static separation are now used to control medium access dynamically!

Multiplexing



Medium Access



Clarification of Terms

- **Static multiplexing** of the resources (frequency, time, space, code)
⇒ Multiplexing schemes (layer 1, see previous chapter)
- **Semi-static reservation** of the resources => this chapter
 - Reservation of a resource for
 - a complete session (i.e. a recurring time slot for a voice call in DECT or GSM) or
 - some smaller time, i.e. to transmit a frame or packet
- **Highly dynamic access** without previous reservation => this chapter
 - Random Access to a resource for some smaller time, e.g. to transmit a single frame, done in a distributed manner

CSMA/CD known from Ethernet as a Solution?

Can we apply (random) media access methods from fixed networks?

Example: CSMA/CD

- **C**arrier **S**ense **M**ultiple **A**ccess with **C**ollision **D**etection (IEEE 802.3)
- send as soon as the medium is free (**carrier sensing – CS**)
- listen to the medium, if a collision occurs stop transmission and jam (**collision detection – CD**)

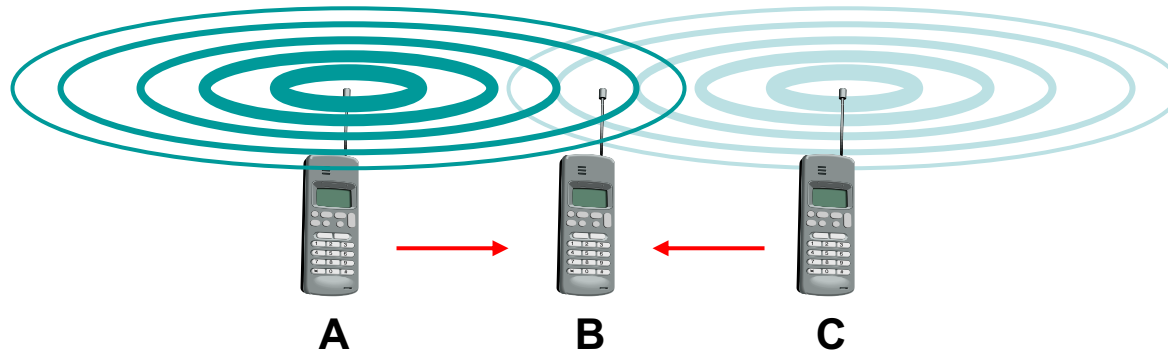
Problems in wireless networks with inconsistent view

The signal strength decreases (at least) proportional to the square of the distance resulting in inconsistent views between sender and receiver:

- the sender would apply CS and CD, but the collisions happen at the receiver
- it might be the case that a sender cannot “hear” the collision, i.e., CD does not work
- furthermore, CS might not work if, e.g., a terminal is “hidden”

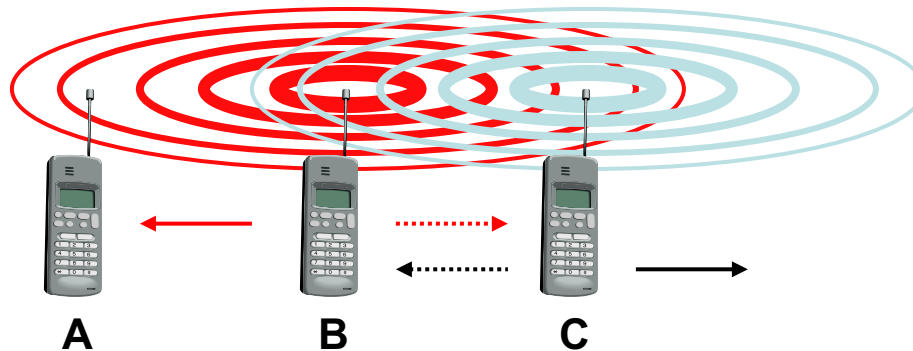
Hidden terminal problem

- A sends to B, C cannot receive A
- C wants to send to B, C senses a “free” medium -> CS fails
- collision at B, A cannot receive C -> CD fails
- A is “hidden” for C



Exposed terminal problem

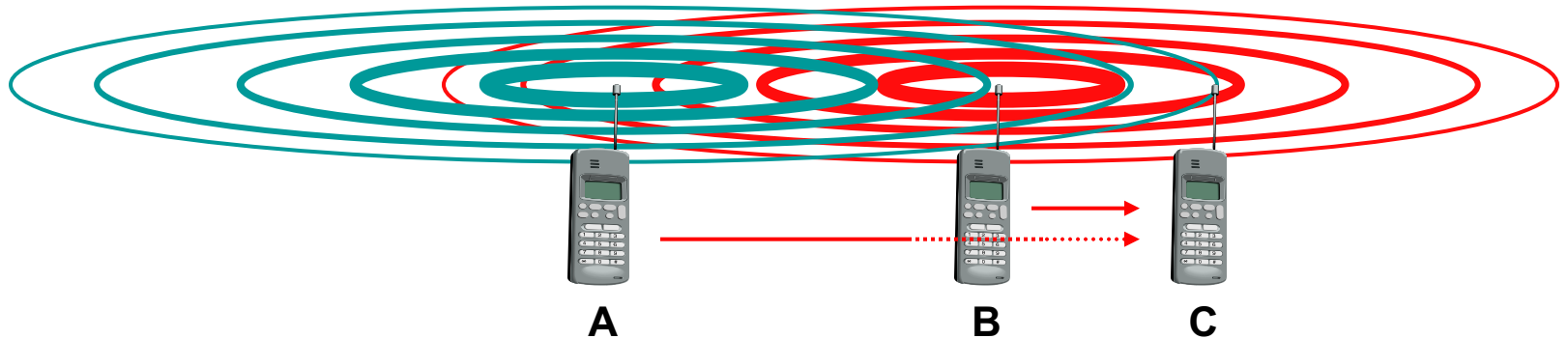
- B sends to A, C wants to send to another terminal (not A or B)
- C has to wait, CS signals a medium in use
- but A is outside the radio range of C, therefore waiting is not necessary
- C is “exposed” to B



Near and far terminal problem

Terminals A and B send, C receives

- signal strength decreases with distance d ($1/d^2 \dots 4$)
- the signal of terminal B therefore drowns out A's signal
- C cannot receive A



If C for example was an arbiter for sending rights, terminal B would drown out terminal A already on the physical layer

Also severe problem for CDMA-networks – precise power control needed!

Multiplexing and Duplexing

Questions to be answered:

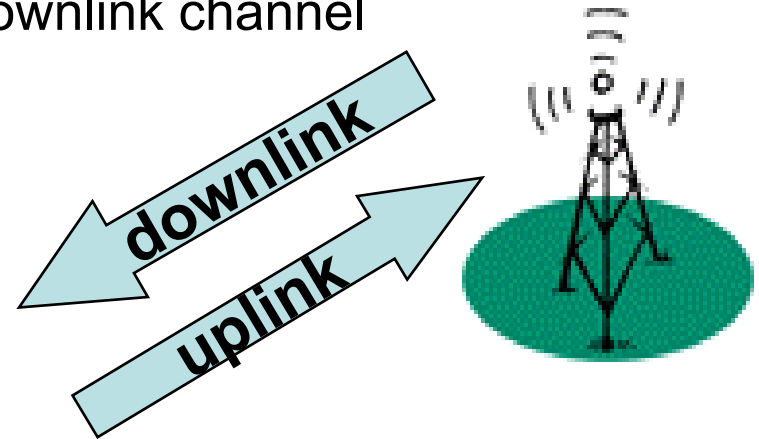
- What is simplex communication? What are example systems for this?
- What are the pros and cons of TDD over FDD?
- For which of the given cases is FDD preferable? Why?
- What does uplink and downlink mean?

Communication link types

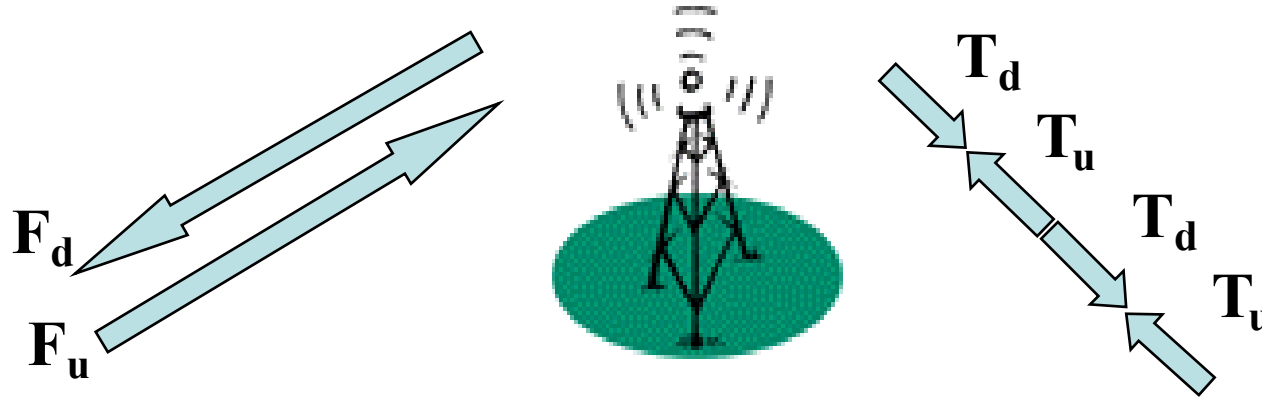
Each terminal needs an uplink and a downlink channel

Types of communication links:

- Simplex
 - unidirectional link transmission
- Half Duplex
 - Bi-directional (but not simultaneous)
- Duplex
 - simultaneous bi-directional link transmission, two types:
 - Frequency division duplexing (FDD)
 - Time division duplexing (TDD)



Duplex modes



Frequency Division Duplex (FDD)

Separate frequency bands for up- and downlink

- + separation of uplink and downlink interference
- no support for asymmetric traffic

Examples: LTE, UMTS, GSM, IS-95, AMPS

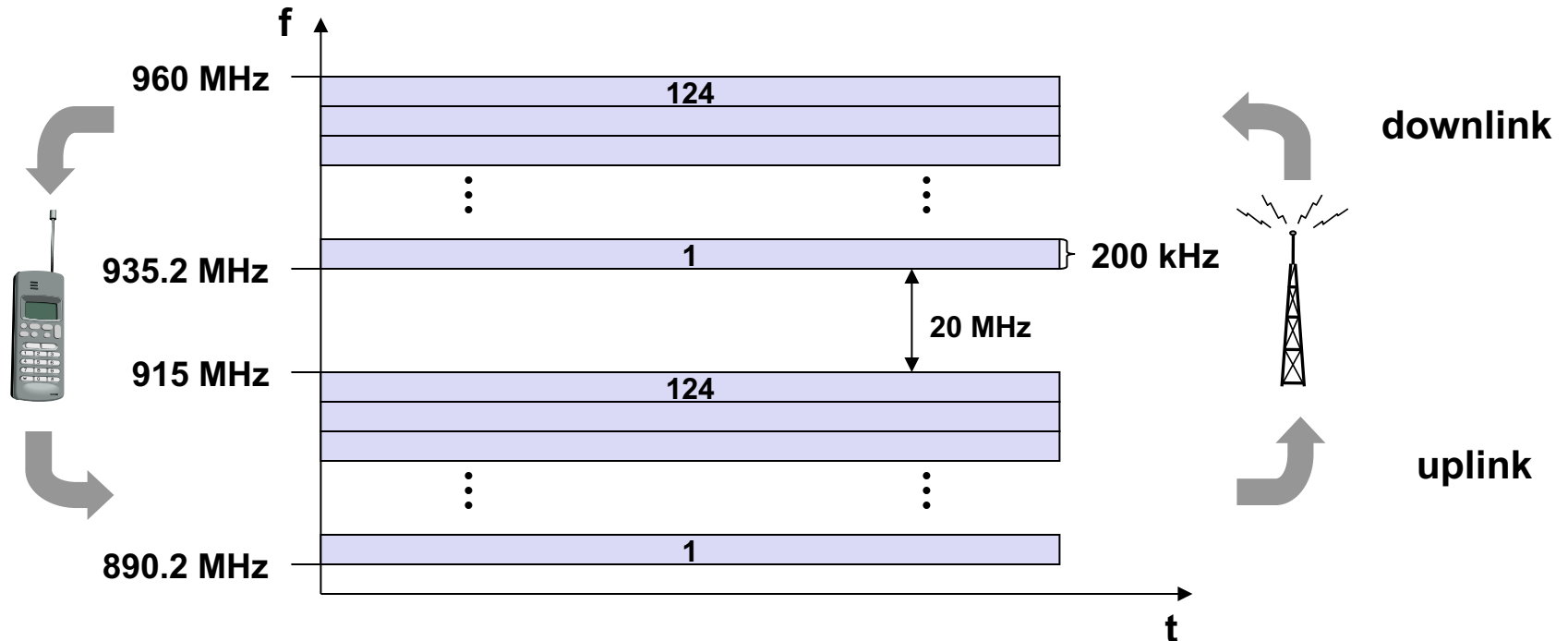
Time Division Duplex (TDD)

Separation of up- and downlink traffic on time axis

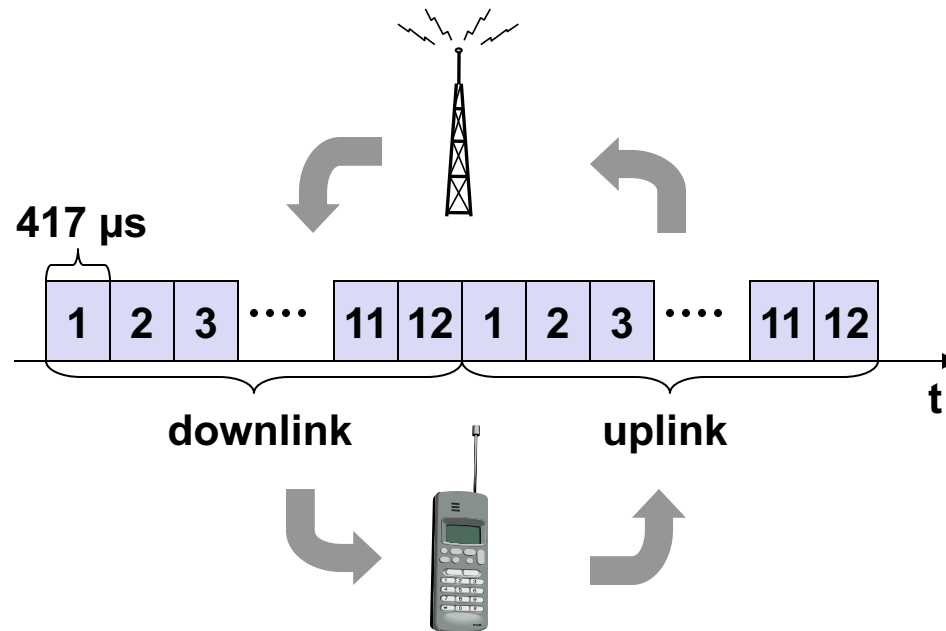
- + support for asymmetric traffic
- mix of uplink and downlink interference on single band

Examples: DECT, WLAN, UMTS (TDD), TD-LTE

FDD/FDMA - general scheme, example GSM



TDD/TDMA - general scheme, example DECT



Time Division Multiple Access

Questions to be answered:

- What is the idea of ALOHA?
- Why does slotted ALOHA outperform ALOHA with respect to throughput?
- What does implicit reservation mean?
- How does explicit reservation work?
- What is the difference between random access and reservation-based access? What are the pros and cons of the two strategies?
- Compare the access latencies of both access schemes. Consider the latency under low and high loads.
- Compare the two access strategies with respect to the maximum load they can handle.
- Name the limits of distributed approaches to media access?
- How can the hidden-terminal problem be solved? Does the RTS/CTS scheme (MACA) solve the problem? In any case?

Time Division Multiple Access

Methods for access coordination:

- **Central coordinator**

- A central entity (base station) coordinates access within a given area in a dynamic (scheduling) or semi-static way (reservations)
- Is applied in all cellular networks (GSM, UMTS, LTE, etc.)
- Scheduling strategies known from operating systems can be applied here
- Mobile station needs to request resources from the base station beforehand (semi-static reservation or dynamic requests)

- **Decentralized coordination => our focus here!**

- Each entity decides on its own when to access, typically based on knowledge acquired from the environment and neighbor nodes
- Risk of access conflicts due to non-consistent views of nodes (e.g. hidden and exposed terminal problems)
- Main access method in WLANs

Time Division Multiple Access

Access strategies

- **Reservation-based access**

- Reservation of the medium for some time and information of potential conflicting partners of the reservation helps to prevent collisions
- Reservations could be long term (for a connection), medium term (for an IP packet or packet burst) or short term (called scheduling)
- Reservation incurs delay and overhead for the reservation but ensures the conflict-free and efficient transmission of data
- Typically a reservation also requires a prior random access to request and reserve the resource (may be centralized or distributed)

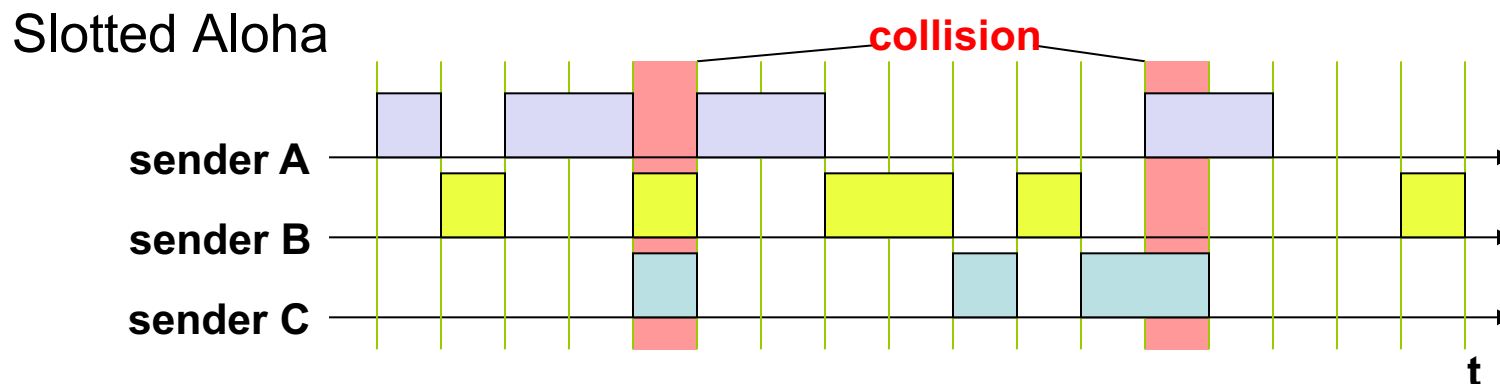
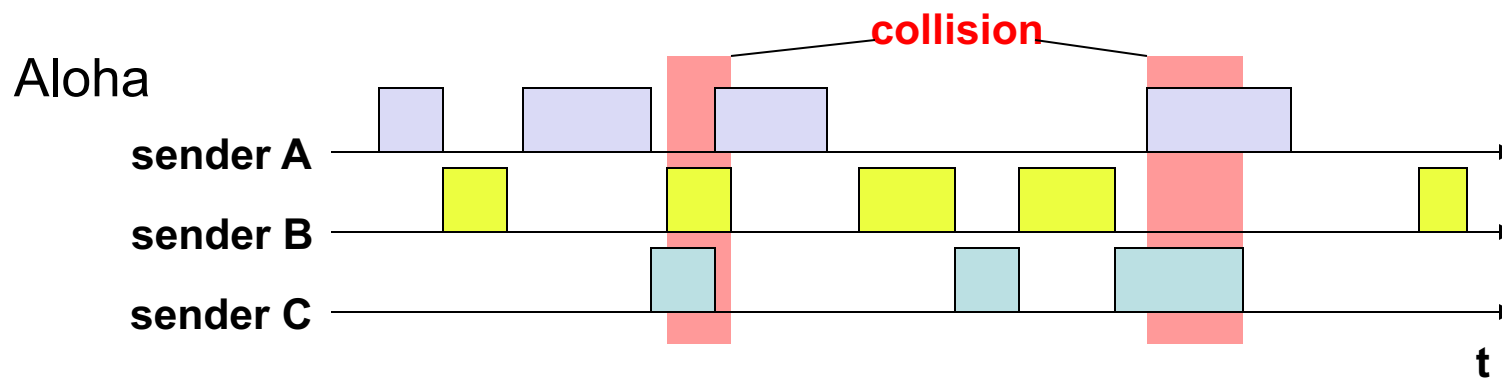
- **(Semi-)Random access**

- First initial access of any new node to the system
- Random access is fast and effective in the case of low resource usage
- Randomness may result in collisions especially if the resource use is high

TDMA: Aloha/slotted aloha

Mechanism

- random, distributed (no central arbiter), time-multiplex
- Slotted Aloha additionally uses time-slots, sending must always start at slot boundaries



TDMA: Demand Assigned Multiple Access (DAMA)

Channel efficiency only 18% for Aloha, 36% for Slotted Aloha (assuming immediate retransmission with probability, Poisson distribution of packet arrivals and packet lengths)

Reservation can increase efficiency close to 100%

Idea:

- a sender *reserves* a future time-slot
- sending within this reserved time-slot is possible without collision

Disadvantage: reservation causes higher delays

Applications:

- typical scheme for satellite links (long round-trip-times)
- application to packet data, e.g. in GPRS and UMTS

Examples for reservation algorithms:

- *Explicit Reservation (Reservation-ALOHA)*
- *Implicit Reservation (PRMA)*
- *Reservation-TDMA*

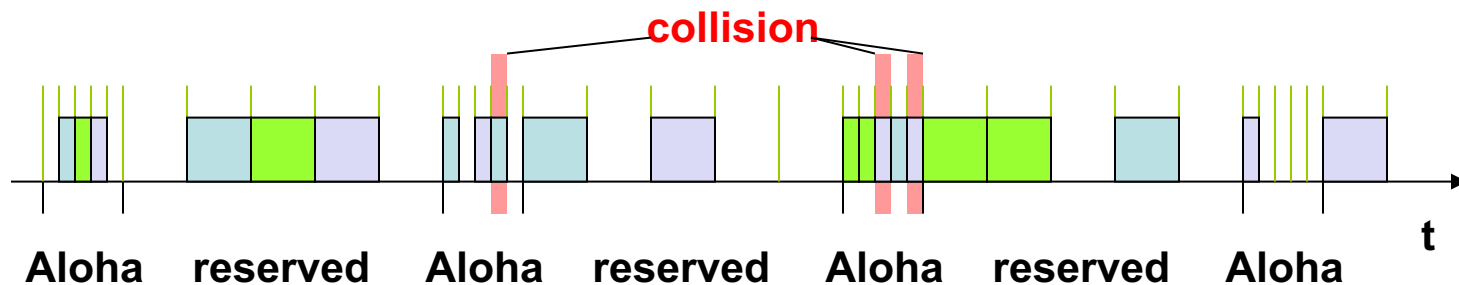
TDMA: DAMA – Explicit Reservation based on Aloha

Explicit Reservation (e.g. applied by Reservation-Aloha):

Two modes:

- *ALOHA mode* for reservation:
competition for small reservation slots, collisions possible
- *reserved mode* for data transmission within successful reserved slots (no collisions)

Synchronisation: it is important for all stations to keep the reservation list consistent at any point in time and, therefore, all stations have to synchronize from time to time

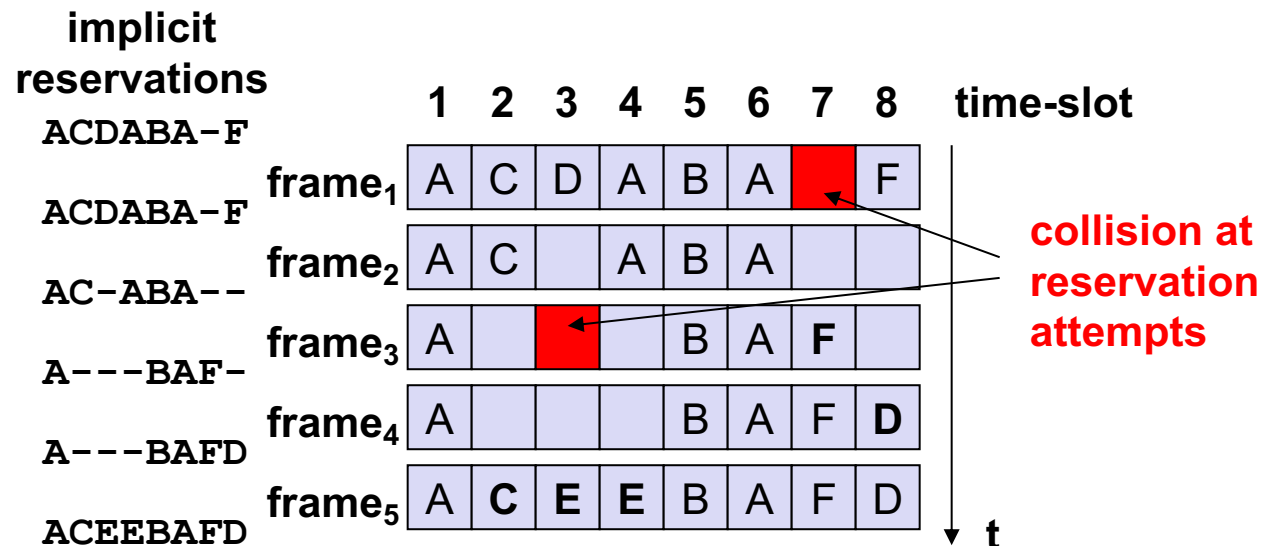


Picture shows a simple variant: different versions exist

TDMA: DAMA – Implicit Reservation

Implicit reservation (PRMA - Packet Reservation MA):

- a certain number of slots form a frame, frames are repeated
- stations compete for empty slots according to the slotted aloha principle
- once a station reserves a slot successfully, this slot is automatically assigned to this station in all following frames as long as the station has data to send (implicit reservation)
- competition for these slots starts again as soon as the slot was empty in the last frame

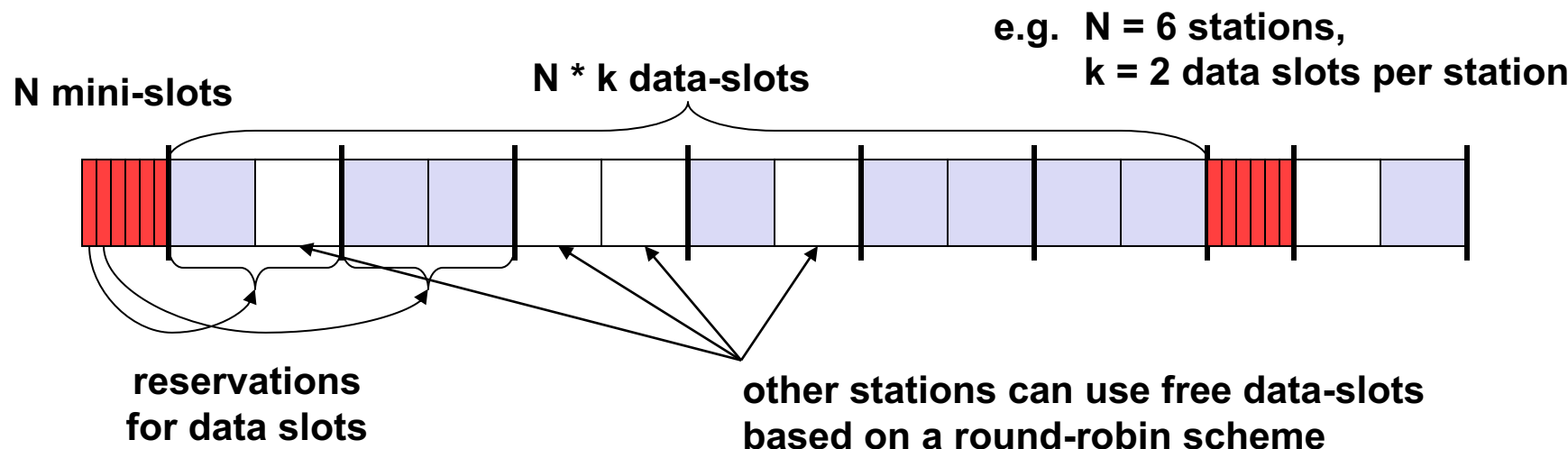


New successful reservation attempts are in bold letters

TDMA: DAMA – Reservation-TDMA

Reservation Time Division Multiple Access

- every frame consists of N mini-slots and x data-slots
- **every station has its own mini-slot** and can reserve up to k data-slots using this mini-slot (i.e. $x = N * k$)
- other stations can send data in unused data-slots according to a round-robin sending scheme (best-effort traffic)



- Advantage: (small) guaranteed bandwidth with small latency for each station
- Disadvantages: fixed number of stations (mini slots); global coordination

TDMA: Multiple Access with Collision Avoidance (MACA)

Motivation: deal with hidden terminals without a base station (central controller)

Idea: use short signaling packets for collision avoidance

- RTS (request to send): a sender requests the right to send from a receiver with a short RTS packet before it sends a data packet
- CTS (clear to send): the receiver grants the right to send as soon as it is ready to receive
- all other stations listen to the signal

Signaling packets contain

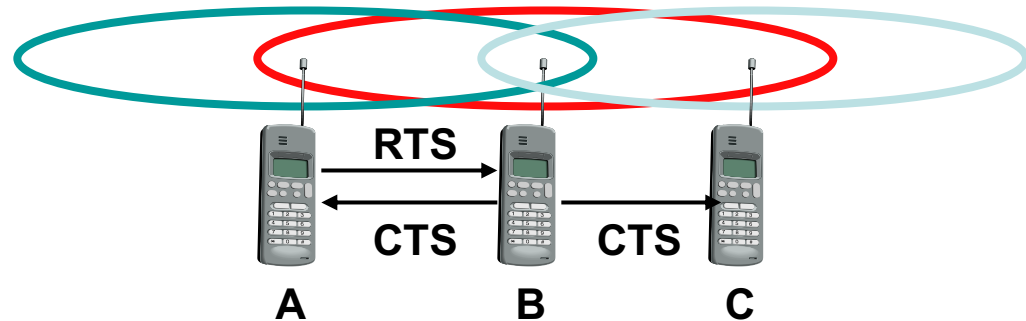
- sender address
- receiver address
- packet size

Collisions are mainly limited to the transmission of RTS signals which is small compared to the data transmission

TDMA: MACA examples

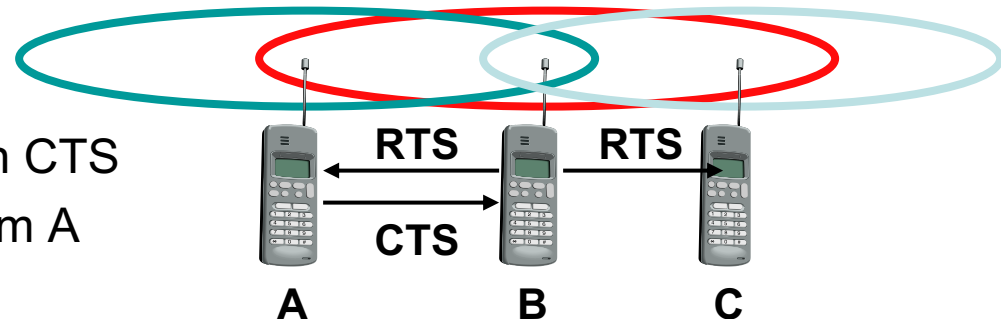
MACA avoids the problem of hidden terminals

- A and C want to send to B
- A sends RTS first
- C waits after receiving CTS from B



MACA avoids the problem of exposed terminals

- B wants to send to A, C to another terminal
- B sends RTS, A replies with CTS
- C does not receive CTS from A
- => C concludes that it is not within receiving range of A
- C can start its transmission



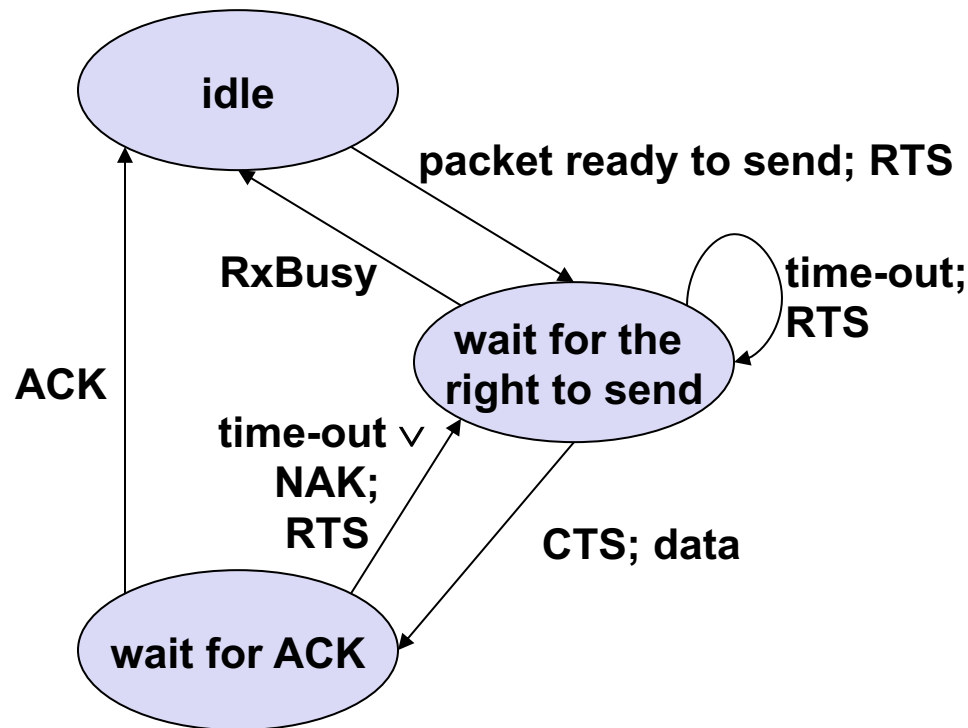
Disadvantage:

- overhead where data packets are small

TDMA: MACA variant: DFWMAC in IEEE 802.11

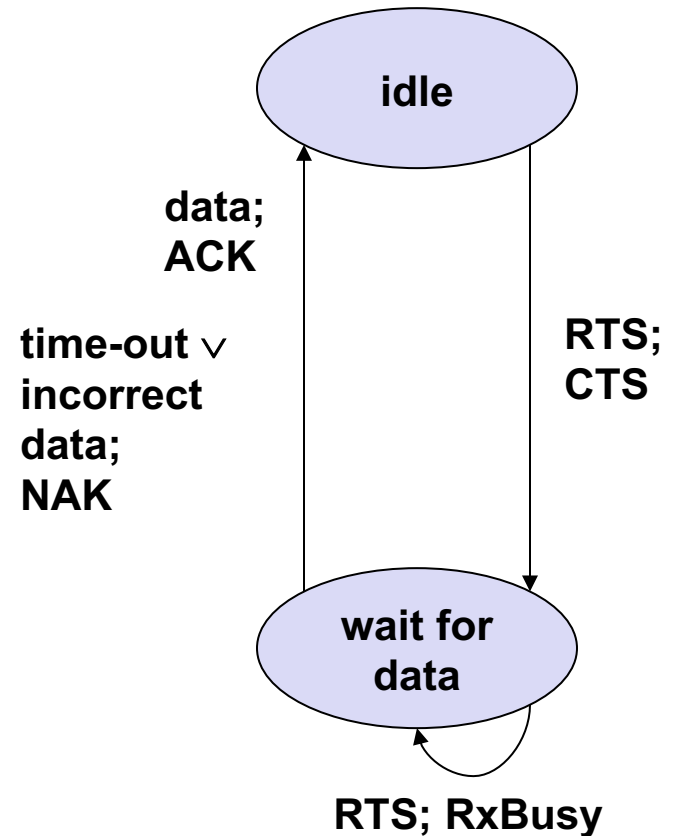
Simplified state machine

sender



ACK: positive acknowledgement
NAK: negative acknowledgement

receiver



RxBusy: receiver busy

TDMA: Polling by central entity

If one terminal can be heard by all others, this “central” terminal (e.g. a base station) can poll all other terminals according to a certain scheme

- now all schemes known from fixed networks can be used (typical mainframe - terminal scenario, round-robin, random, reservation-based)

Example: Randomly Addressed Polling

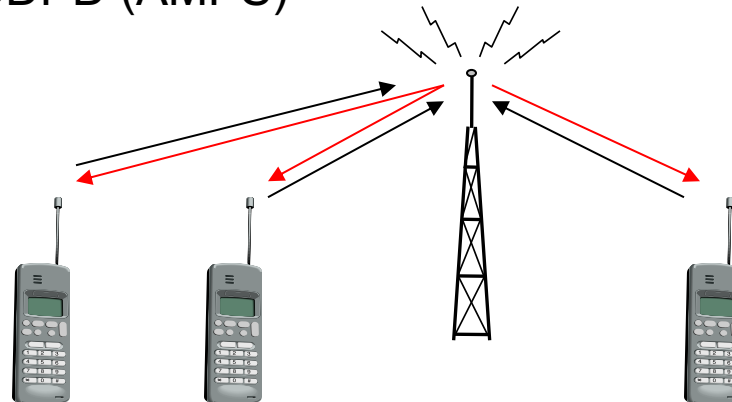
- base station signals readiness to all mobile terminals
- terminals ready to send can now transmit a random number without collision with the help of CDMA or FDMA (the random number can be seen as dynamic address)
- the base station now chooses one address for polling from the list of all received random numbers (collision if two terminals choose the same address)
- the base station acknowledges correct packets and continues polling the next terminal
- this cycle starts again after polling all terminals of the list

Application to Bluetooth and 802.11 (possible access function)

TDMA: ISMA (Inhibit Sense Multiple Access)

Current state of the medium is signaled via a “busy tone”

- the base station signals on the downlink (base station to terminals) if the medium is free or not
- terminals must not send if the medium is busy
- terminals can access the medium as soon as the busy tone stops
- the base station signals collisions and successful transmissions via the busy tone and acknowledgements, respectively (media access is not coordinated within this approach)
- mechanism used, e.g. for CDPD (AMPS)



Code Division Multiple Access

Questions to be answered:

- What is the trick with CDMA to avoid collisions?
- Why is power control important for correct functioning of CDMA?
- What happens if the keys are not orthogonal?
- What is the advantage of CDMA with respect to the handling of noise?
- Does CDMA require the same, higher or lower SINR for correct transmission? Why?

CDMA access method

CDMA (Code Division Multiple Access)

- all terminals send on the same frequency probably at the same time and can use the whole bandwidth of the transmission channel
- each sender has a unique random number, the sender XORs the signal with this random number
- the receiver can “tune” into this signal if it knows the pseudo random number, tuning is done via a correlation function

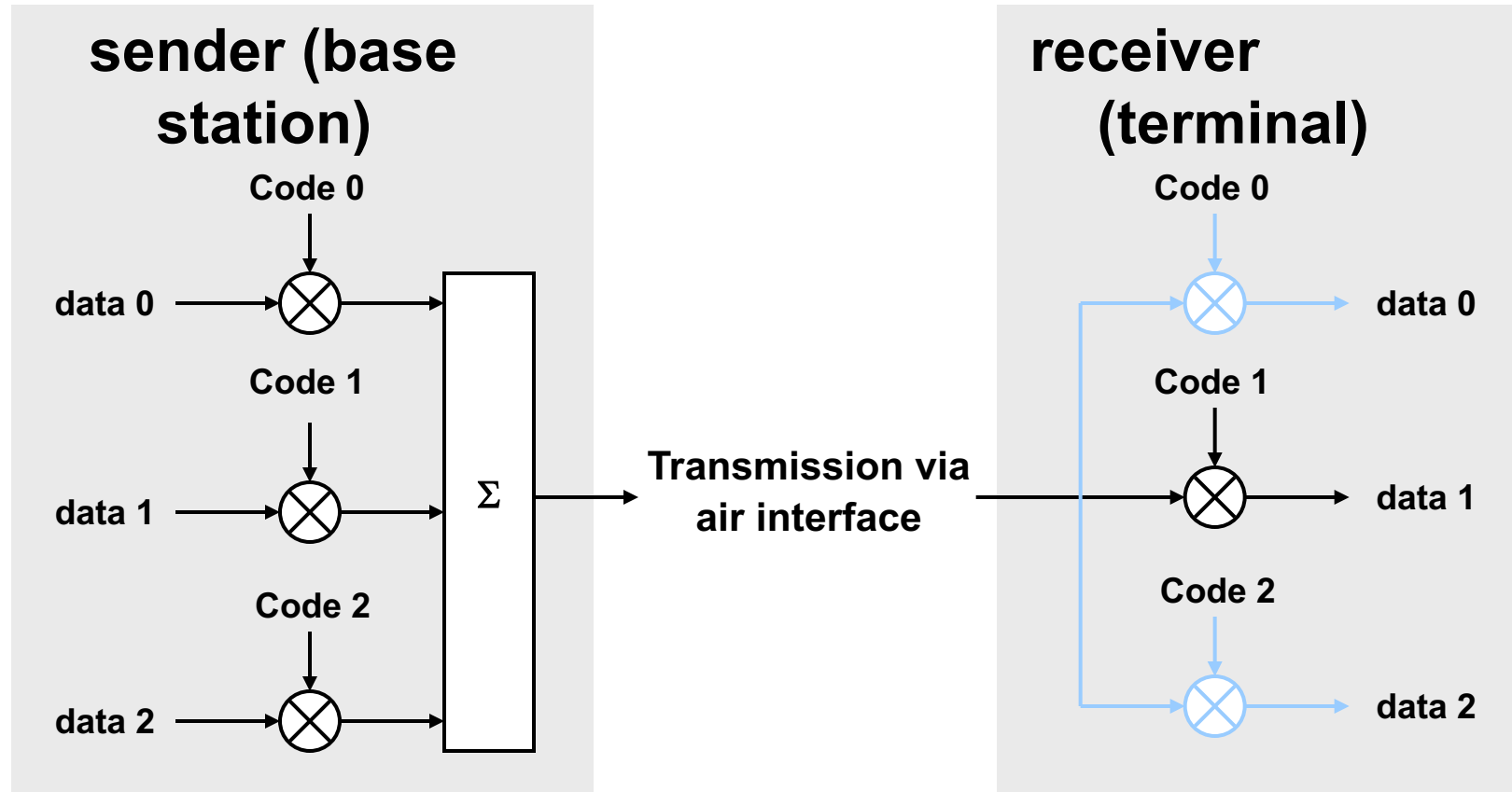
Advantages:

- all terminals can use the same frequency, less planning needed
- huge code space (e.g. 2^{32}) compared to frequency space
- interference (e.g. white noise) is not coded
- forward error correction and encryption can be easily integrated

Disadvantages:

- higher complexity of a receiver (receiver cannot just listen into the medium and start receiving if there is a signal)
- all signals should have the same strength at a receiver (power control)

CDMA principle



CDMA by example

data stream A & B

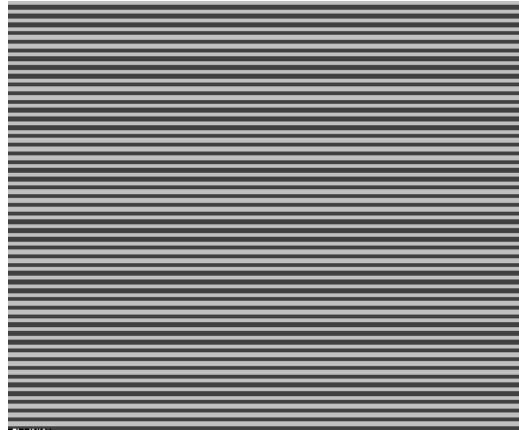
spreading

spreaded signal

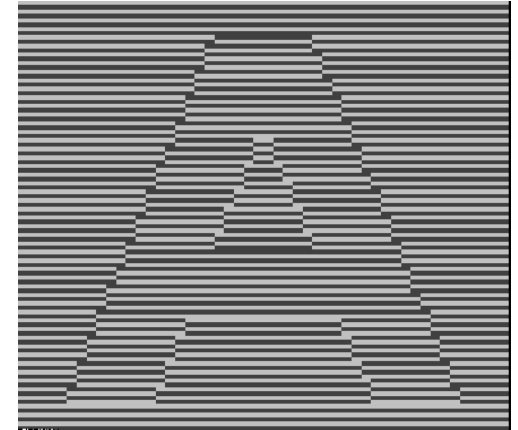
Source 1



Code 1



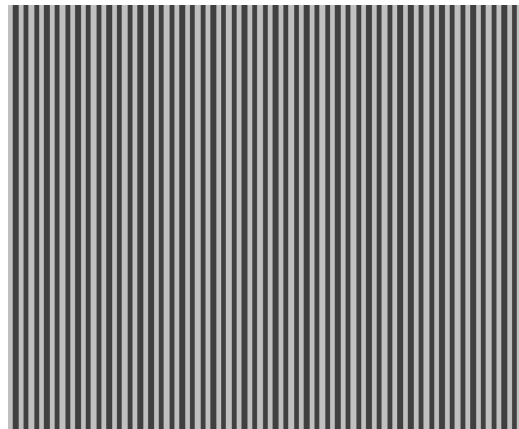
Source 1 spread



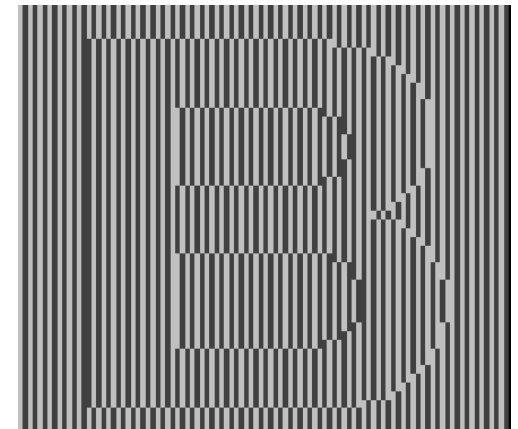
Source 2



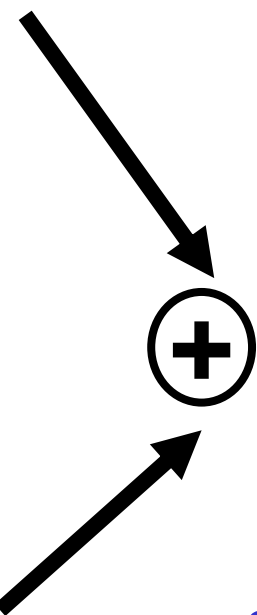
Code 2



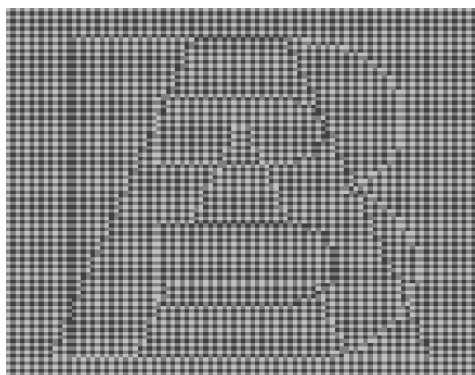
Source 2 spread



CDMA by example

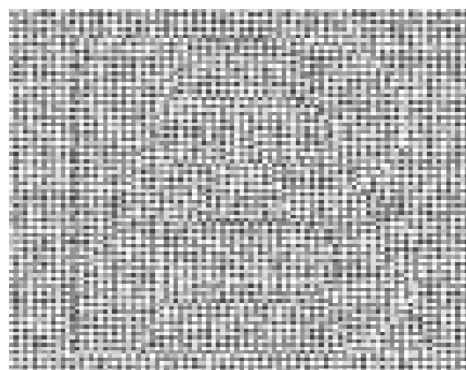


Sum of Sources Spread



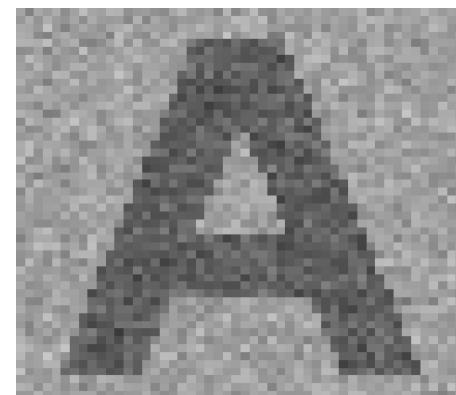
overlay of signals

Sum of Sources Spread + Noise



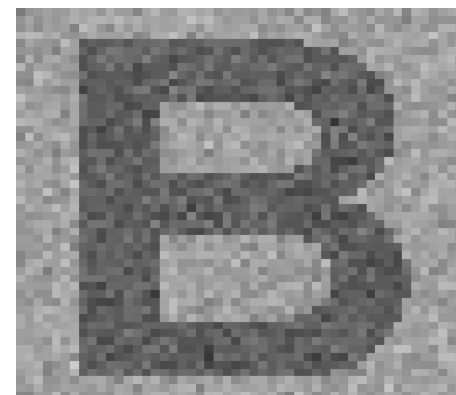
transmission
and distortion
(noise and
interference)

Despread Source 1



decoding
and
despreading

Despread Source 2



See also <http://mkn.dev.geekbetrieb.de/cdma/>

CDMA in theory

Sender A

- sends $A_d = 1$, key $A_k = 010011$ (i.e. -1 1 -1 -1 1 1)
- sending signal $A_s = A_d * A_k = (-1, +1, -1, -1, +1, +1)$

Sender B

- sends $B_d = 0$, key $B_k = 110101$ (i.e. 1 1 -1 1 -1 1)
- sending signal $B_s = B_d * B_k = (-1, -1, +1, -1, +1, -1)$

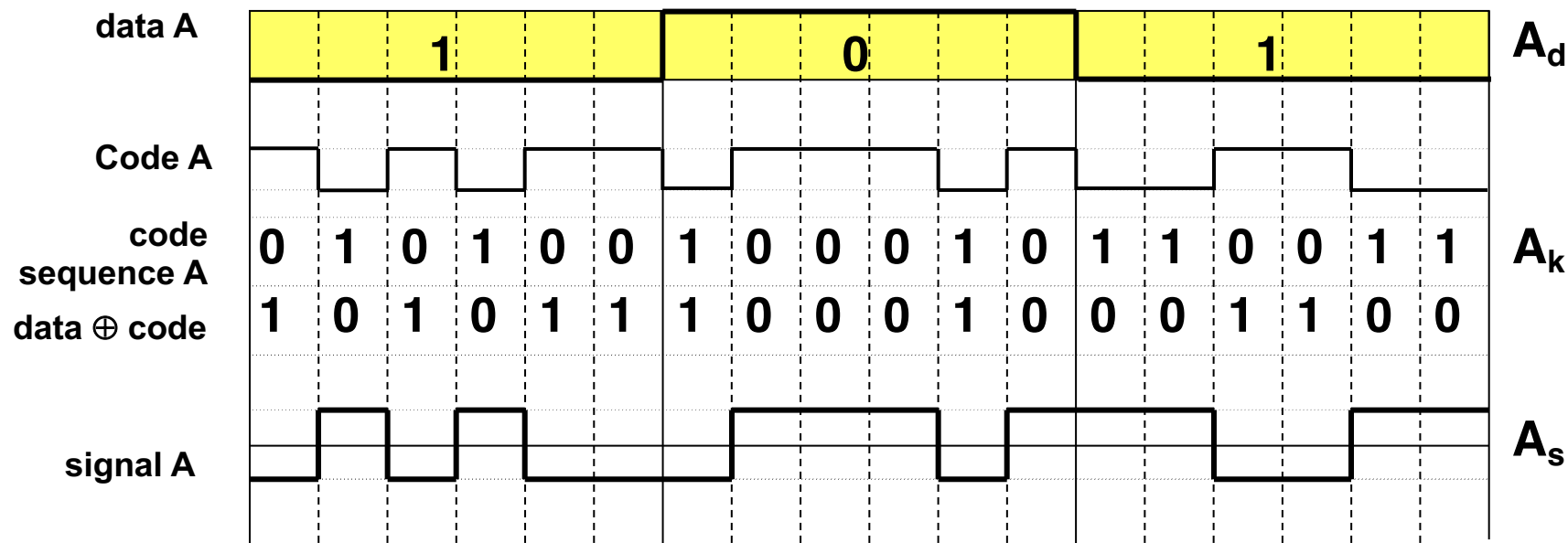
Both signals superimpose in space

- interference neglected (noise etc.)
- $A_s + B_s = (-2, 0, 0, -2, +2, 0)$

Receiver wants to receive signal from sender A

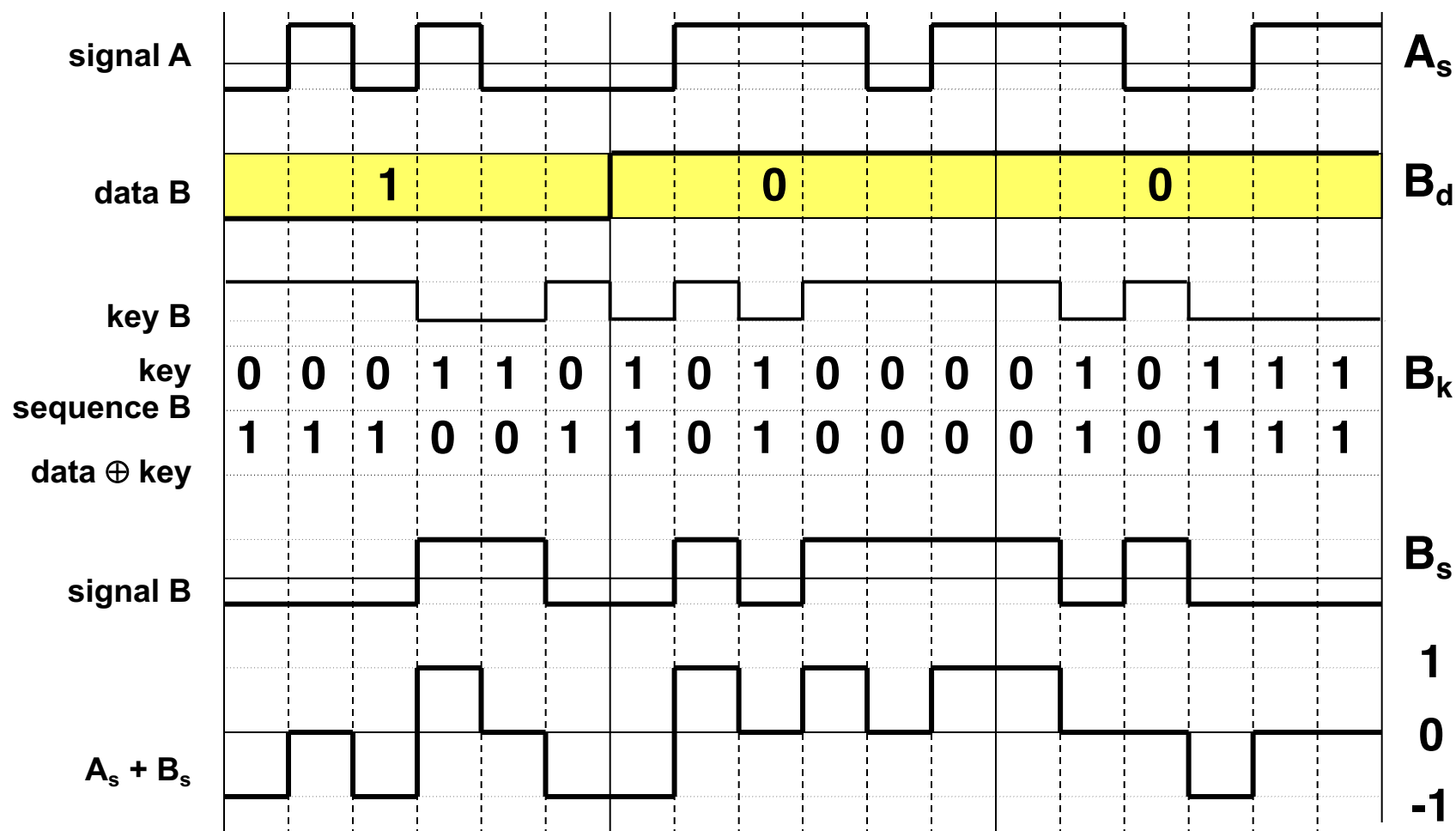
- apply key A_k bitwise (inner product)
 - $A_e = (-2, 0, 0, -2, +2, 0) \bullet A_k = 2 + 0 + 0 + 2 + 2 + 0 = 6$
 - result greater than 0, therefore, original bit was „1“
- receiving B
 - $B_e = (-2, 0, 0, -2, +2, 0) \bullet B_k = -2 + 0 + 0 - 2 - 2 + 0 = -6$, i.e. „0“

CDMA on signal level I

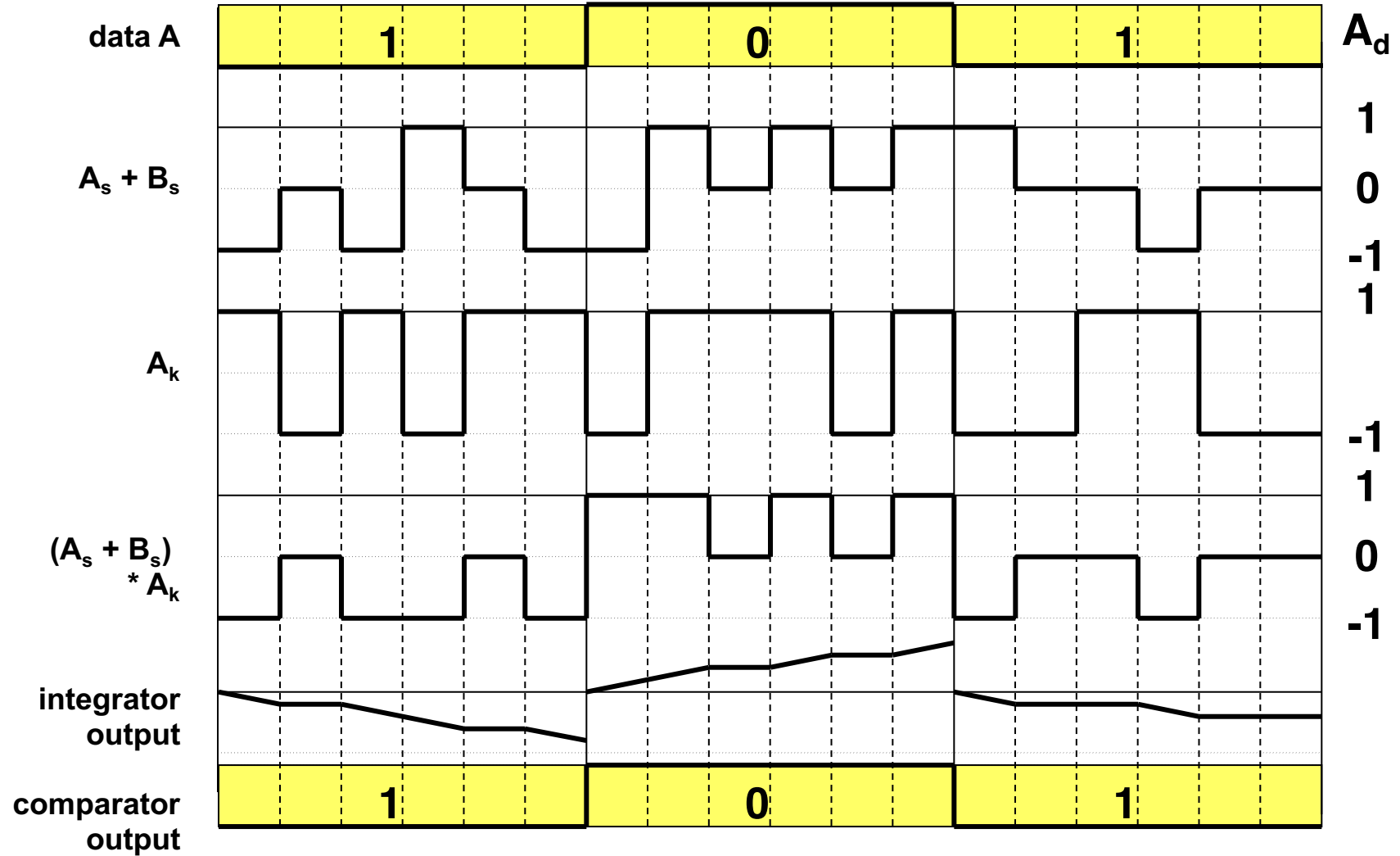


Real systems use much longer keys resulting in a larger distance between single code words in code space

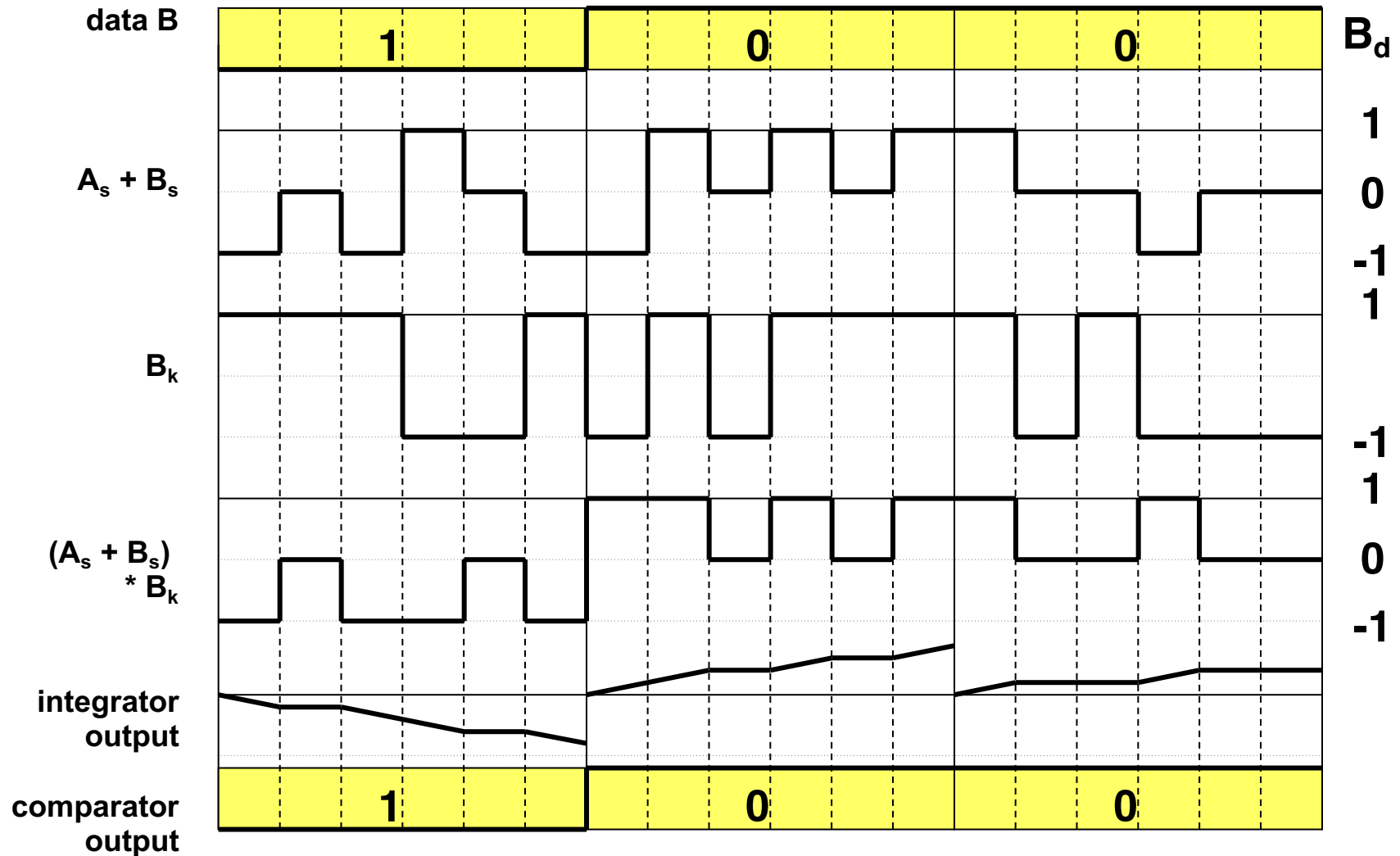
CDMA on signal level II



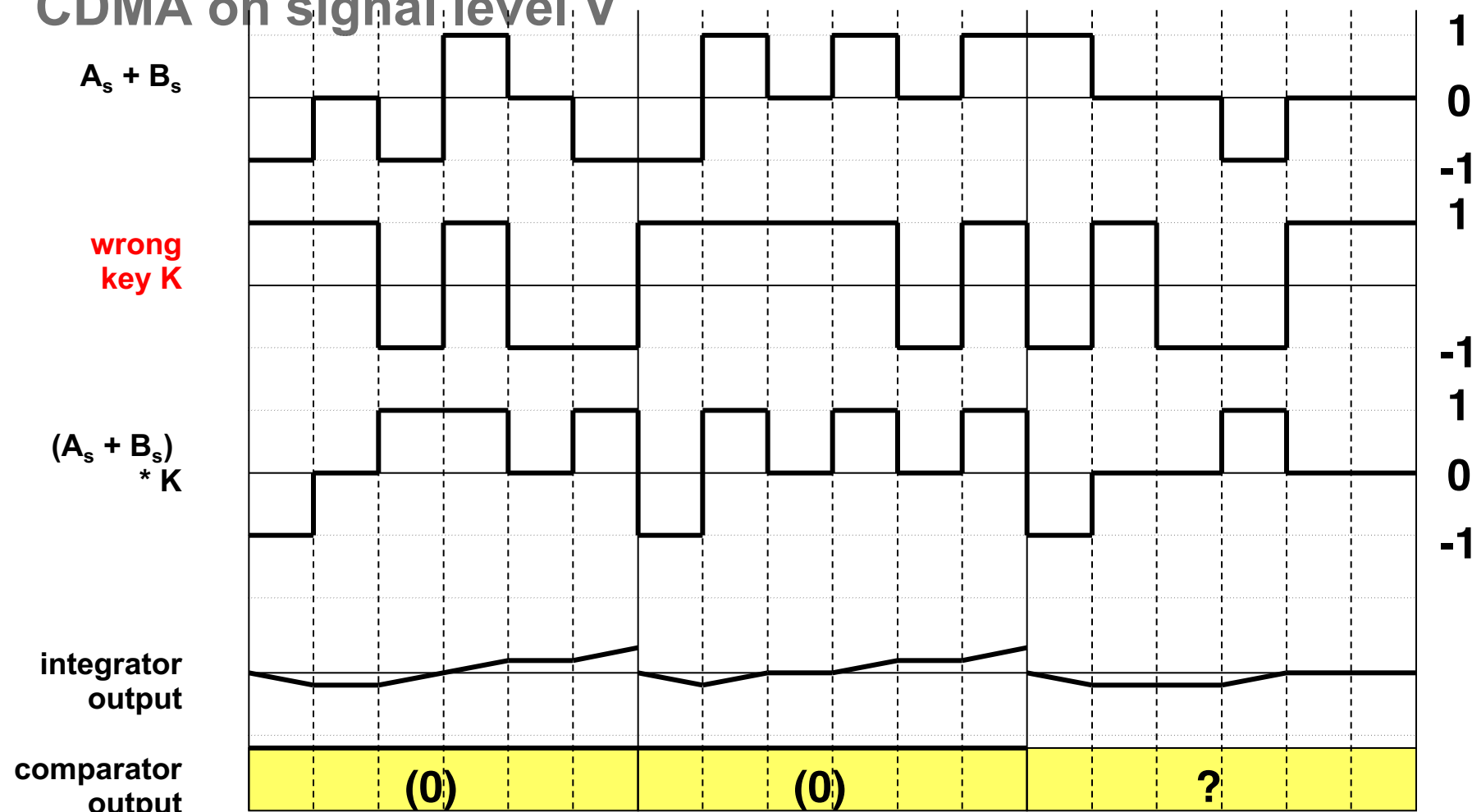
CDMA on signal level III



CDMA on signal level IV



CDMA on signal level V



Assumptions

- orthogonality of keys
- neglectance of noise
- no differences in signal level => precise power control

Comparison SDMA/TDMA/FDMA/CDMA

Approach	SDMA	TDMA	FDMA	CDMA
Idea	segment space into cells/sectors	segment sending time into disjoint time-slots, demand driven or fixed patterns	segment the frequency band into disjoint sub-bands	spread the spectrum using orthogonal codes
Terminals	only one terminal can be active in one cell/one sector	all terminals are active for short periods of time on the same frequency	every terminal has its own frequency, uninterrupted	all terminals can be active at the same place at the same moment, uninterrupted
Signal separation	cell structure, directed antennas	synchronization in the time domain	filtering in the frequency domain	code plus special receivers
Advantages	very simple, increases capacity per km ²	established, fully digital, flexible	simple, established, robust	flexible, less frequency planning needed, soft handover
Dis-advantages	inflexible, antennas typically fixed	guard space needed (multipath propagation), synchronization difficult	inflexible, frequencies are a scarce resource	complex receivers, needs more complicated power control for senders
Comment	only in combination with TDMA, FDMA or CDMA useful	standard in fixed networks, together with FDMA/SDMA used in many mobile networks	typically combined with TDMA (frequency hopping patterns) and SDMA (frequency reuse)	higher complexity, typically integrated with FDMA

Summary

Central vs. decentralized control

- central control:
 - consistent view on resource usage simplifies coordination
 - registration or static configuration of system needed
- decentralized control:
 - disjoint & inconsistent views on shared medium (hidden & exposed terminal)
 - distributed arbitration scheme needed to minimize collisions

Random vs. reserved access

- random access:
 - fast, risk of collisions
- reserved access:
 - slow, high capacity

How to compare access schemes?

- overall throughput/goodput of system
- access latency or packet delay (average or worst case)
- fairness among stations
- QoS guarantees, esp. wrt. delay and throughput

References

Jochen Schiller: Mobile Communications, Addison-Wesley, 2003

(most of the material covered in this chapter is based on the book)

Ramjee Prasad, Marina Ruggieri: Technology Trends in Wireless Communications, Artech House, 2003